Smart Helmet – Component List (Bill of Materials)

S.No	Component Name	Category	Model / Specification	Quantity	Cost
1	ESP32 Dev Board	Microcontroller	ESP32-WROOM-32 (WiFi + BLE)	1	350
2	GSM Module	Communication	SIM800L Quad-Band GSM/GPRS	1	350
3	Alcohol Sensor	Safety Sensor	MQ-3 Alcohol Gas Sensor	1	150
4	Motion Sensor	Crash Detection	MPU6050 Accelerometer + Gyro	1	200
5	Proximity Sensor	Helmet Detection	IR Obstacle Sensor Module	1	50
6	Heart Rate Sensor	Fatigue Monitoring	Pulse Sensor (KY-039)	1	30
7	Buzzer	Alert	Active Buzzer Module (5V)	1	20
8	LEDs	Indicator	Red LED + Green LED (5mm)	2	20
9	Battery	Power	Li-ion 18650 – 3.7V	1	80
10	Charging Module	Power Management	TP4056 with Protection	1	80
11	Boost Converter	Voltage Regulation	MT3608 DC-DC Step-Up	1	50
12	Power Switch	Control	Mini Slide / Toggle Switch	1	10
13	Jumper Wires	Connectivity	Male–Female	AS required	100

14	Prototype Board	Assembly	General Purpose PCB / Breadboard	1	100
15	Enclosure	Mechanical	Helmet Mount Enclosure	1	300